29th February 2024

Financial Analytics Lab

GROUP 11

- Prithvi Raj Suryavanshi 20CH3FP53
- Vinay Siwach 20CH3FP49
- Shrit Gautam 20CH3FP26
- Sanyam Agrawal

02 # Answers

[Link to PYTHON CODE]

Q1. We have chosen 'Amount of Prepaid Card($)' as the target variable.

- Correlation of features with the target variable:
- Age: 0.215141
- Days per month at Starbucks 0.406864
- Cups of Coffee per Day: 0.286227
- Income ($1000): 0.850032

- **Income Dependency:** A noteworthy observation lies in the significant correlation between affluent individuals, indicated by a high income coefficient of 0.85, and their propensity to utilize prepaid cards. This suggests a strong inclination for those with greater financial means to opt for prepaid card solutions.
- **Starbucks Patronage and Expenditure:** Further analysis unveils a conspicuous relationship between the frequency of visits to Starbucks establishments and the utilization of prepaid cards. Evidently, individuals who patronize Starbucks frequently exhibit a marked preference for prepaid cards, presumably enticed by associated rewards or incentives offered by such cards.
- **Coffee Consumption Dynamics:** Intriguingly, there exists a discernible correlation between individuals' daily coffee intake and their adoption of prepaid cards, albeit to a lesser extent compared to other factors. This implies that individuals with higher coffee consumption may find value in employing prepaid cards for their transactions, albeit not as prominently as other observed correlations.
- **Age Stratification Analysis:** Although age alone does not emerge as the predominant determinant, disaggregating data into distinct age cohorts yields insightful nuances. Such segmentation promises a deeper understanding of the varying motivations and drivers behind prepaid card usage across different age demographics.

Answers

Implications for Sales Strategy

- **Targeting Affluent Demographics:** Given the robust association between income levels and prepaid card utilization, strategically directing marketing efforts towards high-income individuals holds promise for maximizing sales opportunities.

- **Leveraging Starbucks Affiliation:** Capitalizing on the evident link between Starbucks patronage and prepaid card usage presents a prime opportunity to augment sales. Emphasizing the advantages of prepaid cards, such as rewards accrual, for Starbucks transactions can effectively engage and incentivize loyal patrons.

- **Tailored Marketing Strategies by Age:** Crafting customized marketing campaigns tailored to specific age brackets enables a more nuanced approach to address divergent motivations for prepaid card adoption. Understanding the distinct preferences and requirements of each age demographic is essential for optimizing marketing effectiveness.

- **Cultural Alignment with Coffee Consumption:** Recognizing the inherent association between coffee culture and prepaid card usage, devising promotional initiatives that underscore the convenience and benefits of prepaid cards for avid coffee enthusiasts can resonate effectively within this niche segment.

- **Incentivizing Younger Demographics:** Despite a relatively weaker correlation, targeting younger age groups, particularly millennials and Gen Z, with targeted incentives and rewards aligned with their lifestyle preferences can serve to stimulate uptake and usage of prepaid cards within these cohorts.

Answers

Q2.
Model Strength:
The model exhibits a moderate explanatory power with an R-squared of 0.416 and an adjusted R-squared of 0.332. The F-statistic (4.984) is significant at p = 0.00913, indicating overall model significance. However, the non-significant coefficients for x1 (Age) and x3 suggest that Age and another predictor may not meaningfully contribute to the model.

Predictor Correlations:

Age: The correlation with the target variable, Days per month at Starbucks, is only 0.037457. Age doesn't strongly influence Starbucks visits, making our marketing age-neutral.

Cups of Coffee per Day: With a high correlation of 0.587601, coffee consumption strongly correlates with prepaid card usage. This suggests an opportunity to target coffee enthusiasts by emphasizing the convenience of prepaid cards for their daily caffeine fix.

Income ($1000): Income has a moderate correlation (0.305438) with the target variable. While it matters, it's not as crucial. Our marketing strategy should be tailored to middle-income individuals, showcasing how prepaid cards make everyday spending easier and more rewarding.

Market Analysis and Conclusions:

Targeting Coffee Enthusiasts: Given the strong correlation between Cups of Coffee per Day and Starbucks visits, focusing on coffee enthusiasts is a viable strategy. Emphasizing the convenience of prepaid cards for daily coffee purchases could attract this demographic.

Income Considerations: While income plays a role, it's not the primary driver. Middle-income individuals are a key market segment to target, showcasing the practical benefits of prepaid cards for everyday spending.

Age-Neutral Marketing: Since Age has a minimal impact on Starbucks visits, keeping marketing age-neutral is recommended. Highlighting the universal benefits and convenience of prepaid cards can appeal to a broader audience.

Model Limitations: The non-significant coefficients for Age and another predictor in the model suggest that these factors may not significantly contribute to Starbucks visits. Further investigation into individual predictors is advised for a more comprehensive understanding.

In conclusion, a strategic marketing approach targeting coffee enthusiasts, middle-income individuals, and employing age-neutral messaging can capitalize on the insights derived from the model.

Answers

Q3.
How Strong is the model:
R2 value: 0.999779745985403
this implies that our model demonstrates perfect fit

coefficent of respective features are:
NO. OF STORES : -0.02642496722952925
NO. OF DRINKS : -75.20448212078583
AVG WEEKLY EARNINGS : 38.98929209463208
Hence, No. of Drinks and Avg. Weekly Earnings are significant predictors.

The analysis offers Starbucks management valuable insights into the drivers behind sales revenue. It highlights "NO. OF DRINKS" and "AVG WEEKLY EARNINGS" as key factors that strongly influence revenue. The findings suggest that boosting drink sales and catering to customers with higher weekly earnings could be instrumental in boosting overall revenue. With meaningful coefficients and a robust R-squared value indicating a well-fitting model, this analysis equips management with actionable intelligence to refine sales tactics and predict future revenue trends effectively.

Feature Coefficients:

- No. of Stores: The coefficient is -0.02642496722952925, suggesting a negative relationship with the target variable. However, its statistical significance is not mentioned.
- No. of Drinks: The coefficient is -75.20448212078583, indicating a significant negative impact on the target variable. Each unit increase in the number of drinks is associated with a decrease in the target variable.
- Avg. Weekly Earnings: The coefficient is 38.98929209463208, suggesting a positive impact on the target variable. This predictor is statistically significant, and an increase in average weekly earnings is associated with an increase in the target variable.

Market Analysis and Conclusions:

- Perfect Fit: The near-perfect fit of the model raises questions about overfitting. While a high R-squared value is desirable, caution is needed to ensure the model generalizes well to new data.
- Significant Predictors:
  - No. of Drinks: The negative coefficient implies that a higher number of drinks is associated with a decrease in the target variable. This could indicate potential market saturation or other factors affecting consumer behavior.
  - Avg. Weekly Earnings: The positive coefficient indicates a positive impact on the target variable. This suggests that higher average weekly earnings correlate with an increase in the target variable, providing an avenue for market growth.
- No. of Stores: The negative coefficient suggests a potential negative impact on the target variable, but the statistical significance is not provided. Further investigation is needed to determine the significance of this predictor and its contribution to the model.
- Caution on Overfitting: The extremely high R-squared value raises concerns about overfitting, where the model may be too tailored to the training data. Validation on new data is crucial to ensure the model's generalizability.